DAY 1 • 60 MIN

Data Drift & Unsupervised Anomaly Detection

Building smoke alarms so a machine learning model never fails silently in production.

TODAY'S OUTCOMES

- Explain why a deployed model quietly loses accuracy over time.
- Tell apart Feature Drift, Concept Drift, and Label Drift.
- Use the Population Stability Index (PSI) to prove data has shifted.
- Train an Isolation Forest to catch weird records with no labels.
- Sketch a simple monitoring architecture for production models.

MENTAL MODEL

A model is a snapshot.
The world keeps moving.

Monitoring is how you find out your snapshot is out of date — before your users do.

01 • THE CORE PROBLEM

Why models decay in production

You train on a snapshot of the world. The world doesn't hold still.

> *“A model that was 95% accurate at training can silently decay to 60% accuracy within 3 months in production.”*

ROOT CAUSE

The real world changes, but the model's learned rules don't update themselves.

COMMON TRIGGERS

- Seasonality & economic shifts
- Market crashes or crises
- New regulations
- New customer segments
- Sensor failures
- Data pipeline bugs

THE SNAPSHOT ANALOGY

Imagine training a model is like taking a photo of a crowd. Months later, the photo still shows the same faces — but the real crowd has changed: new people, new behavior, new context.

A model that keeps trusting the photo will keep making decisions for a crowd that no longer exists.

Three ways data can drift

Same word, three different failure modes — each needs a different response.

TYPE	WHAT CHANGES	EXAMPLE
Feature Drift	Input distribution $P(X)$	Customer demographics shift after a pandemic
Concept Drift	Relationship $P(Y\|X)$	Old signals stop predicting churn as behavior changes
Label Drift	Output distribution $P(Y)$	Fraud rate doubles during an economic crisis

PSI: putting a number on drift

Population Stability Index — the industry-standard drift score, especially in banking.

THE FORMULA

```
PSI = Σ (Actual% − Expected%) × ln(Actual% / Expected%)
```

Split data into buckets, compare the % of records in each bucket between a reference period and today.

WHY BANKS USE IT

It works on any feature (income, age, credit score), gives one clean number per feature, and is cheap to run daily across hundreds of columns.

THRESHOLDS

< 0.1

Stable

Monitor normally

0.1 – 0.2

Slight shift

Investigate the feature

> 0.2

Major shift

Retrain the model

Two more tools worth knowing

PSI isn't the only way to test whether two distributions differ.

KS TEST

Kolmogorov–Smirnov Test

A statistical test that compares two distributions directly and returns a p-value.

```
if p_value < 0.05:
    print("DRIFT DETECTED")
```

Best for: numeric features, when you want a rigorous statistical yes/no answer.

CHI-SQUARE TEST

For Categorical Features

Compares category counts (e.g. plan type, region) between two time periods.

```
chi2, p, dof, exp =
  chi2_contingency(table)
```

Best for: categorical columns where PSI/KS don't apply directly.

Isolation Forest: catching the one weird point

Drift affects a whole dataset. Anomalies are single, impossible data points.

THE FOREST ANALOGY

Imagine you're lost in a forest.

Stand next to a common pine tree, and it takes many random cuts to fence you off from all the other pine trees around you.

Stand next to one neon-pink glowing tree, and a single random cut isolates you immediately.

Isolation Forest draws random “cuts” through the data. Points that take very few cuts to isolate are flagged as anomalies.

WHY “UNSUPERVISED”?

It never needs labeled examples of “normal” vs “anomaly.” It just finds points that are structurally easy to separate from the rest — perfect for real-time production monitoring.

KEY PARAMETERS

contamination Expected anomaly rate (e.g. 5%)

n_estimators Number of random trees

Output 1 = normal, −1 = anomaly

A simple monitoring architecture

Drift checks and anomaly checks both feed the same alert-and-retrain loop.

Production Traffic → Feature Store / Logging Layer → Drift & Anomaly Monitor (PSI, KS, Isolation Forest) → **Alert: Investigate → Retrain?**

MONITORING CADENCE

< 10K records / day	**Weekly batch report**
10K – 1M records / day	**Daily batch report**
> 1M records / day	**Hourly streaming monitor**

THE LOOP, IN WORDS

Every batch of production data is checked for drift (PSI/KS) and scanned for anomalies (Isolation Forest). When a threshold is crossed, the data science team investigates and decides whether to retrain and deploy a new model version.

06 • SUMMARY

Key takeaways

Carry these five ideas into any production ML system.

No model lives forever

A deployed model without monitoring is a ticking time bomb.

Feature drift ≠ concept drift

Different causes need different responses.

PSI > 0.2 is the alarm bell

The universal banking-industry signal to retrain.

Isolation Forest needs no labels

Perfect for real-time production monitoring.

QUIZ — CHECK YOUR UNDERSTANDING

1. A global pandemic changes how people spend money overnight. Is this Feature Drift or Concept Drift?
2. If your PSI score for “age” is 0.05, should you be worried?
3. Why is an Isolation Forest considered an “unsupervised” algorithm?
4. What is a “reference window” in drift monitoring?